

DIGITAL IMAGE PROCESSING

LECTURE # 14

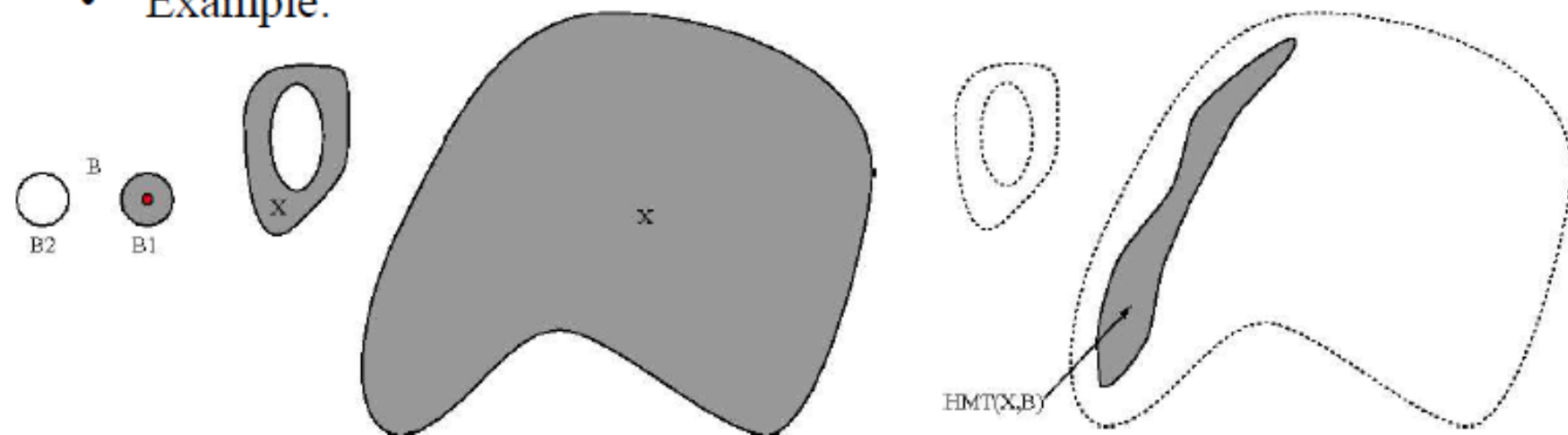
MORPHOLOGICAL IMAGE PROCESSING-II

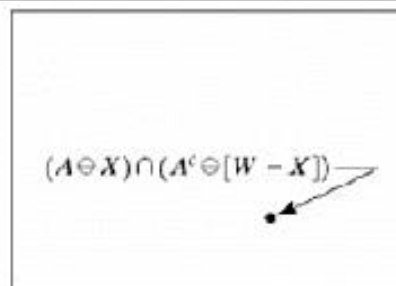
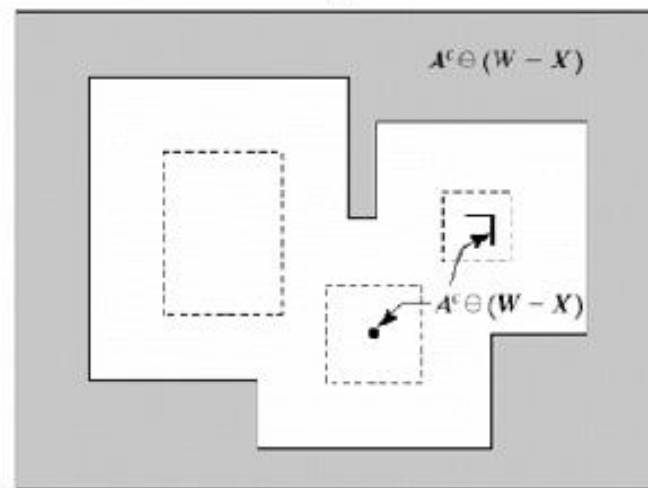
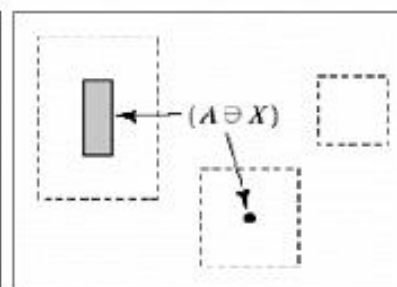
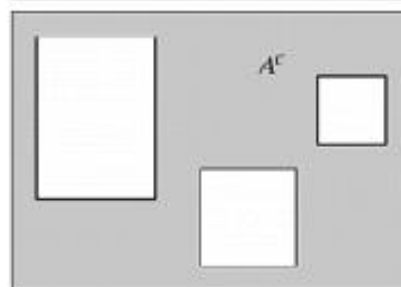
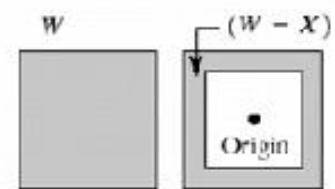
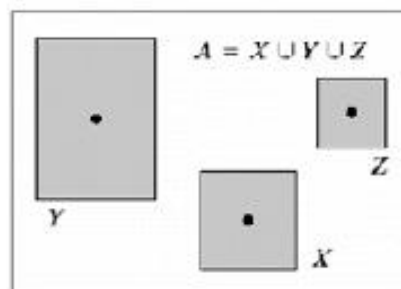
Hit-or-miss transform

- Used to extract pixels with specific neighbourhood configurations from an image
- Grey scale extension exist
- Uses two structure elements B1 and B2 to find a given foreground and background configuration, respectively

$$HMT_B(X) = \{x | (B_1)_x \subseteq X, (B_2)_x \subseteq X^c\}$$

- Example:





Hit or Miss Transform

- Objective is to find a disjoint region (set) in an image
- If B denotes the set composed of X and its background, the match/hit (or set of matches/hits) of B in A , is

$$A \circledast B = (A \ominus X) \cap [A^c \ominus (W - X)]$$

- Generalized notation: $B = (B_1, B_2)$
 - B_1 : Set formed from elements of B associated with an object
 - B_2 : Set formed from elements of B associated with the corresponding background

[Preceeding discussion: $B_1 = X$ and $B_2 = (W - X)$]

- More general definition:

$$A \circledast B = (A \ominus B_1) \cap [A^c \ominus B_2]$$

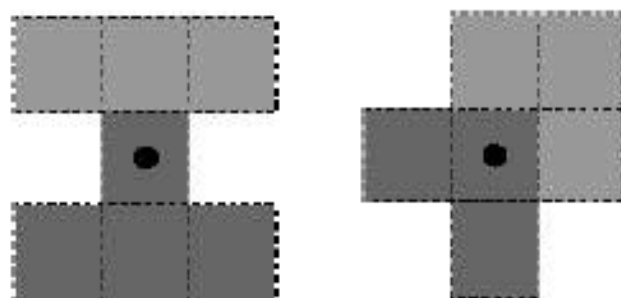
- $A \circledast B$ contains all the origin points at which, simultaneously, B_1 found a hit in A and B_2 found a hit in A^c

Thinning

- Used to shrink objects in binary images
- Differs from erosion in that objects are never completely removed
 - Preserving the homotopy is often an objective
- Successive thinning until stability results in object skeletons
- Thinning is defined as:

$$A \otimes B = A - (A \circledast B) = A \cap (A \circledast B)^c$$

- Example structure elements used in thinning (rotated 90 degrees 3 times to create 8 structure elements):



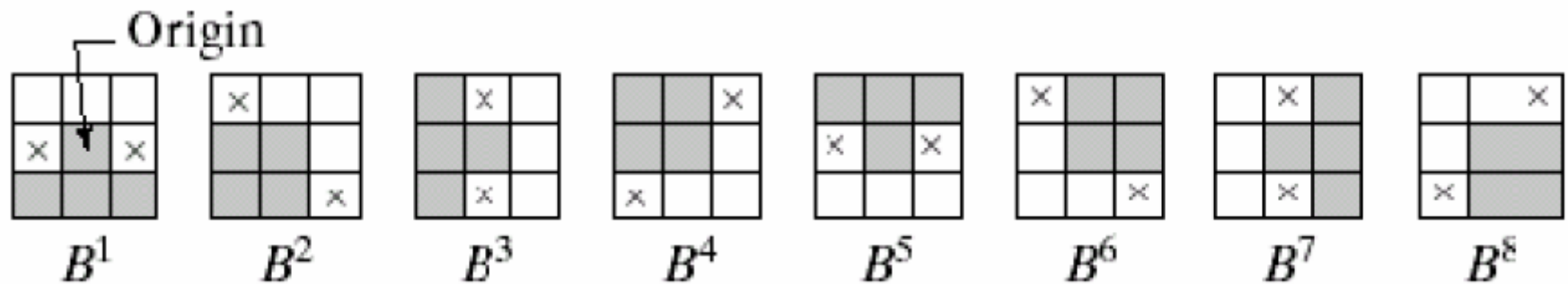
Symmetric thinning

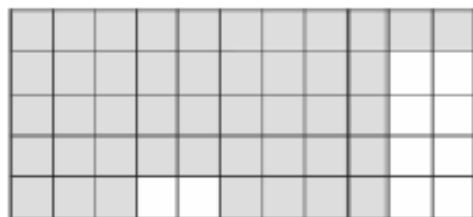
- Sequence of structuring elements,

$$A \otimes \{B\} = (((\dots ((A \otimes B^1) \otimes B^2) \dots) \otimes B^n)$$

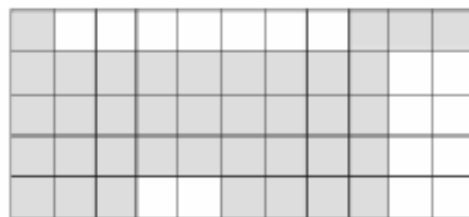
$$\{B\} = \{B^1, B^2, B^3, \dots, B^n\},$$

where B^i is a rotated version of B^{i-1}

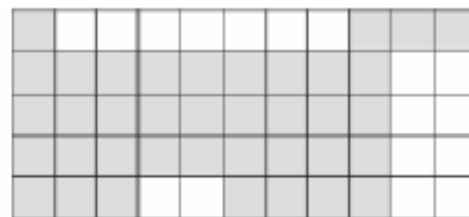




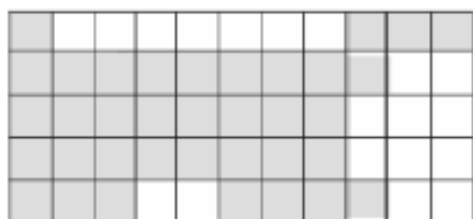
A



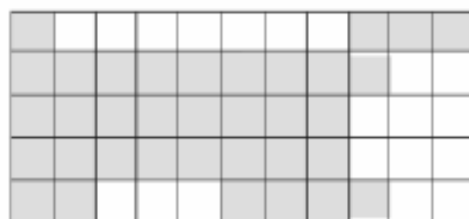
$A_1 = A \otimes B^1$



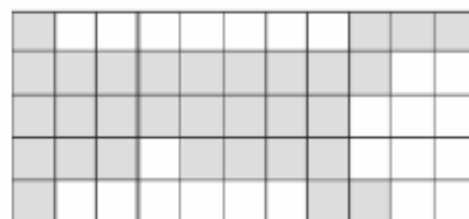
$A_2 = A_1 \otimes B^2$



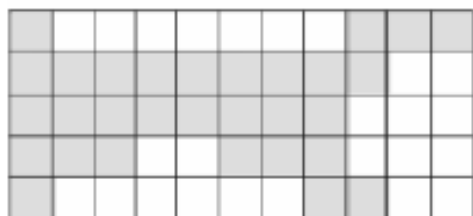
$A_3 = A_2 \otimes B^3$



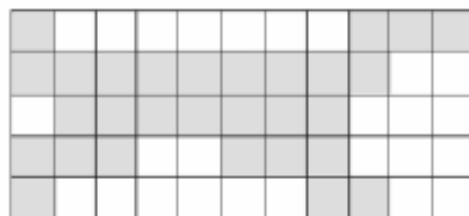
$A_4 = A_3 \otimes B^4$



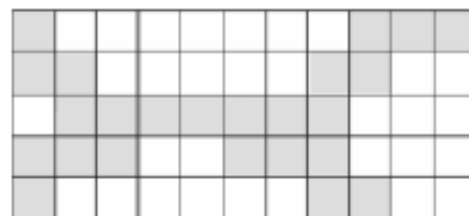
$A_5 = A_4 \otimes B^5$



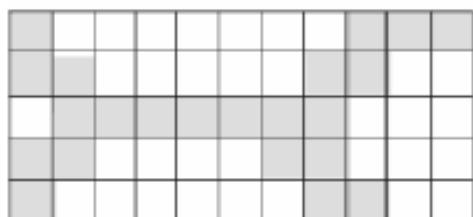
$A_6 = A_5 \otimes B^6$



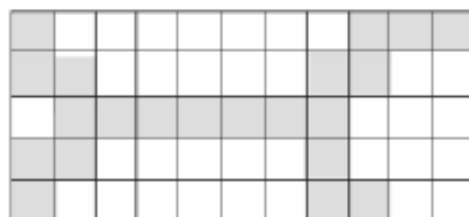
$A_8 = A_6 \otimes B^{7,8}$



$A_{8,4} = A_8 \otimes B^{1,2,3,4}$

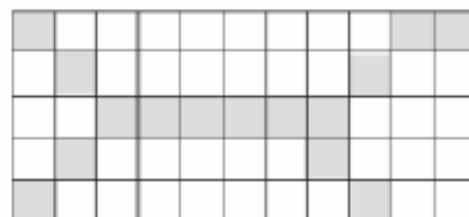


$A_{8,5} = A_{8,4} \otimes B^5$



$A_{8,6} = A_{8,5} \otimes B^6$

No further changes after this.



$A_{8,6}$ converted to m -connectivity.

Thickening

Thickening is the morphological dual of thinning and is defined by

$$A \odot B = A \cup (A * B),$$

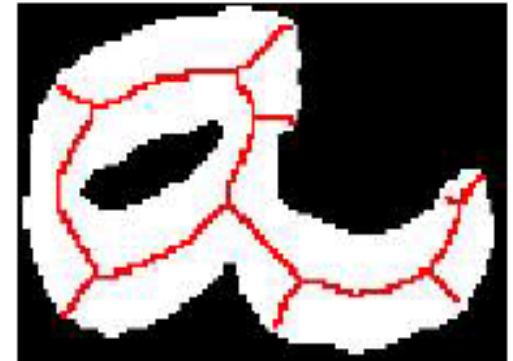
where B is a structuring element

Similar to thinning...

$$A \odot \{B\} = (((\dots ((A \odot B^1) \odot B^2) \dots) \odot B^n)$$

Skeletons / Medial Axis

- Compact or minimal representation of objects in an image while retaining homotopy of the image
- As stated earlier, the skeletons of objects in an image can be found by successive thinning until stability
- The thinning cannot be executed in parallel since this may cause the homotopy of the image to change
- Example:



Skeletons / Medial Axis cont.

- The skeleton of an object is often defined as the medial axis of that object.
 - Pixels are then defined to be skeleton pixels if they have more than one “closest neighbours”.
- Some skeleton algorithms are based on this definition and are computed through the distance transform
- Other algorithms produce skeletons that are smaller than the defined medial axis (such as minimal skeletons)

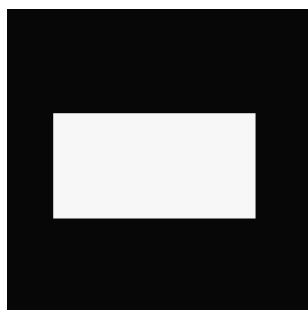
Minimal Representation of Skeleton

- Problem: Finding a minimal representation
 - Solution ■ Pruning of smaller branches
 - Can use HMT to locate and remove endpoints successively

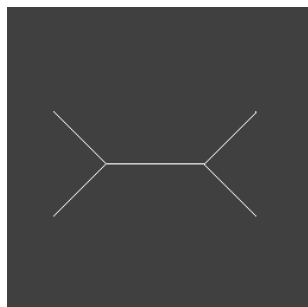


Skeleton with unwanted branches

Skeletonization



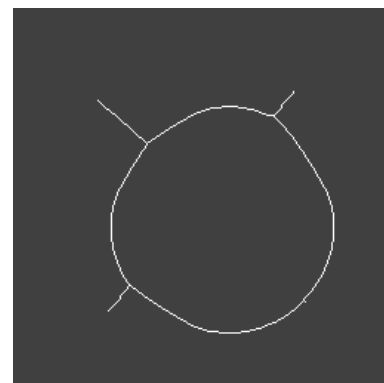
Original Image



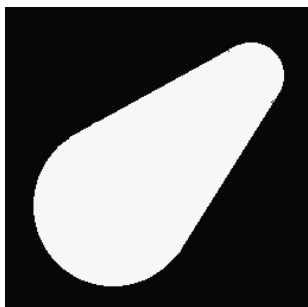
Skeleton



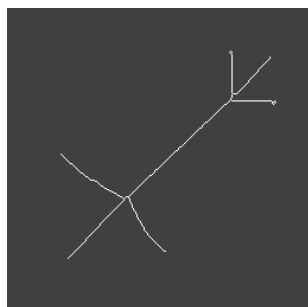
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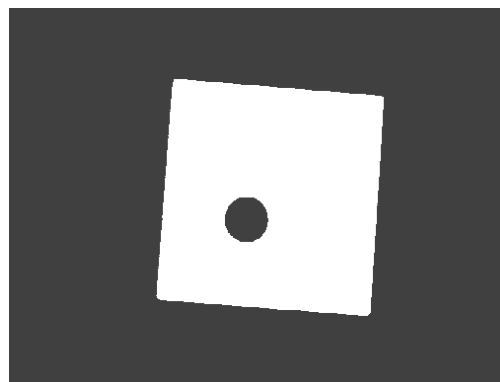
Skeleton



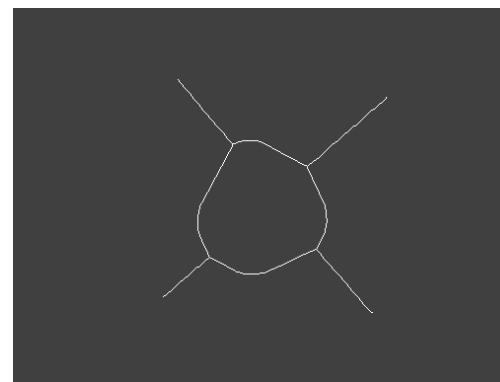
Original Image



Skeleton



Original Image



Skeleton



For any query Feel Free to ask

